



Umut Uygur

Computer & ML Engineer

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OBJECTIVE AND INTERESTED TOPICS

Graduated Computer Engineering student with strong interest in Data science, Machine Learning, Artificial Intelligence, and software development. Focus on building practical systems that solve real-world problems using machine learning and computer vision. Currently developing a computer vision-based monitoring system for the university library. Participates in Kaggle competitions and data-driven projects, including pile load experiment data analysis. Develops thermal image processing applications on edge devices within the SAYZEK graduation project. Develops software primarily using Java, Python, C and SQL. Interested in AI applications, data-driven systems, and full-stack development. Seeks opportunities to contribute to innovative projects in Artificial Intelligence and Data Science while continuously improving technical skills .

PROFESSIONAL EXPERIENCE

- Intern – R&D Software (Image Processing)**

Havelsan Teknoloji Radar (HTR) • 11.08.2025 – 20.09.2025

 - Worked on real-time multi-object tracking and re-identification across cameras using YOLO, BoT-SORT, ReID, MPT, MOT, Torch, OSNet, and implemented cross camera tracking, GStreamer pipelines, WebSocket communication, and retracking algorithms.
- Project Researcher & Project Manager – innovationNTogether Program**

Nurul Teknoloji • 20.10.2025 – 20.06.2026

 - Taking technical and project management responsibilities in the development of an AI-powered Demand Forecasting and Inventory Optimization system, including ERP/WMS data analysis, LightGBM forecasting, Safety Stock, ROP and EOQ algorithms, FastAPI services, sprint planning, and coordination.
- Software Engineer**

ETKİM • 08.2025 - 05.2026

 - Developing an educational VR project supported by the Turkish Ministry of National Education (MEB), focusing on immersive 3D simulations. Responsible for scenario design, 3D environments, user interactions, and VR development within the Etkim Incubation Program..
- Computer Engineering Intern**

TARAFTech GmbH • Germany • 02.2026 - 04.2026

 - Working as a Full Stack Intern with a focus on web-related projects, contributing to the design, development, and implementation of web applications while gaining hands-on experience in full stack development.
- AI Engineer Intern**

Microsoft • 05.2026 - 07.2026

 - Developed a multilingual Retrieval-Augmented Generation (RAG) system during the Microsoft AI Engineer Internship, integrating document retrieval, semantic search, and LLM-based response generation while gaining hands-on experience with AI, and scalable software development.
- Artificial Intelligence Intern**

ULAK Haberleşme A.Ş. • 03.08.2026 – 18.09.2026

 - Developing an enterprise RAG assistant within the Artificial Intelligence and Information Systems Directorate using Self-RAG, Corrective RAG, ChromaDB, local LLMs, Ollama, hybrid BM25/dense retrieval, Cross-Encoder reranking, RRF, query rewriting, conversation memory, OCR, and document processing, with Flask, React, and TypeScript used for application integration.

EDUCATION

- Computer Engineering (English)**

TED University

 - Ankara, Turkey 2022- Present (%50 scholarship, GPA: 3.39)

ADDITIONAL EXPERIENCE

- President**

TEDU IEEE • 2024 – 2027

 - Oversaw IEEE student activities and initiatives. Board Member IEEE CS Committee(2024 – 2025)
- Camp Management / Worker(SWEDEN)**

Herrang dance camp • 2023 – 2025

 - Experienced in ethics, teamwork, problem-solving, and working under pressure with intercultural cooperation..
- President, Founder and Instructor**

TEDU Partner Dances Society • 2022 - 2025

 - Led society activities and instructed members.
- IBM SkillsBuild student ambassador**

IBM • 2026 – 2027

CERTIFICATES

- Data science & Artificial Intelligence Summer School**

Coderspace
- Google Project Management**

Google
- HKUST Software Engineering Specialization**

HKUST
- IEEE CSCON'25 Education Certificate**

IEEE
- Huawei Cloud Certified Developer Associate - AI (HCCDA-AI)**

Huawei
- Milli Teknoloji Akademisi Artificial Intelligence Specialist**

STB
- IBM – Generative AI & Data Science Essentials | LLMs, Data Analysis, Classification, Big Data & Visualization**

IBM

KEY SKILLS

Programming: Python, Java , SQL , HTML/CSS , C , JavaScript , C# , TypeScript **Technologies:** Machine Learning , Data Science , Data Analysis , Computer Vision , YOLO , Multi-Object Tracking , Generative AI , Large Language Models (LLMs) , Data Classification , Full-Stack Development , Web Development , Virtual Reality (VR) , Project Management , Team Collaboration , PyTorch , OpenCV , Linux , TensorFlow

PROJECTS

Multi-Camera Real-Time Object Tracking & Re-Identification System with YOLO + ReID -HTR Jul 2025 – Sep 2025

- B uilt a **real-time multi-camera tracking system (15 live cameras)** that **re-identifies** a target within **2s** across cameras even without visible faces, includes **<1s sabotage alarm detection**; fully operational and presented to **HAVELSAN R&D engineers**, receiving strong **recognition**.
- **Technologies:** YOLOv11s, TorchReID (OSNet), BoT-SORT, Python, GStreamer, Flask, WebSocket, Multi-Object Tracking, Re-Identification, Real- Time AI

Library occupancy detector and system with computer vision-TED University Library Jan 2025 – Jan 2026

- Developed a **computer vision** occupancy monitoring system covering **2 university libraries (40+ cameras, 400 seats)** performing **real-time empty/occupied/reserved seat detection**, implemented with **105,000 TL approved institutional funding**.
- **Technologies:** Python, OpenCV, YOLOv8 (Ultralytics), NumPy

Hybrid Intelligence for Deep Semantic Analysis of UAV Thermal Imagery in Closed Networks Sep 2025 – Present

- Defense-oriented research project analyzing **UAV thermal** imagery using **fine-tuned VLM models** and **edge** deployment in **closed networks**, conducted under **SAYZEK** with planned academic publication.
- **Technologies:** Python, PyTorch, Vision-Language Models (VLM), Thermal Image Processing, OpenCV, Edge AI, Model Fine-Tuning, NumPy, Linux

Values & Career Simulator (Unity, OpenXR) – VR project for Turkish Ministry of Education Aug 2025 – Present

- Immersive VR simulation for student career awareness and decision-making, built in **ODTÜ Teknokent** under the leadership of **TEDU Education Dean**
- **Technologies:** Unity 6.1, OpenXR, C#, XR Origin

Real-World Pile Load Test Data Project – TED University Nov 2025 – Present

- Applying **data science** methodologies to real pile load test datasets from **large-scale** construction projects (malls, skyscrapers) provided by **Dr. Cem Akgüner (TED University)**.
- **Technologies:** Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn.

AI-Powered Demand Forecasting & Inventory Optimization System-Nurol Teknoloji 10/2025

- Developed an AI-based demand forecasting and inventory optimization system using **industrial supply-chain data**, designed for enterprise workflows under **senior AI and software engineer** mentorship
- **Technologies:** Python, LightGBM, Pandas, NumPy, Scikit-learn, FastAPI, PostgreSQL, SQLAlchemy, Docker, React/Streamlit, APScheduler, Alembic, Agile.

Full functional LMS with database 08/2023

- Built a **secure** 3-role LMS with full university database integration, replicating real LMS functionality and receiving full marks in Advanced Database course led by **Dr. Kasım Murat Karakaya (TEDU)**.
- **Technologies:** Java ,Swing , MySQL

LANGUAGES

English (Advanced), Turkish (Native), German (Basic)

REFERENCE

Kutay Öksüz from Havelsan Teknoloji Radar

Dr. Ali Berkol from TEDU and BITES

Dr. Eren Ulu from TEDU

Dr. Evren Coskun from TEDU

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